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Paper 1: Population PK–PD Modeling of Danicopan Add-On Therapy in Participants With Parox

Bibliographic Information

- **Title:** Population PK–PD Modeling of Danicopan Add-On Therapy in Participants With Paroxysmal Nocturnal Hemoglobinuria Treated With Ravulizumab or Eculizumab
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Quick Take

This population PK-PD analysis of danicopan, a factor D inhibitor, in healthy volunteers and PNH patients on C5 inhibitors demonstrates that the approved 150 mg and 200 mg three-times-daily regimens achieve near-complete alternative pathway inhibition regardless of food intake. Clinicians and pharmacometricians should read this to understand the basis for the dosing recommendations and the minimal impact of covariates like body weight, sex, and renal impairment on clinical efficacy.

Executive Summary

Danicopan is a first-in-class oral factor D inhibitor approved as add-on therapy to ravulizumab or eculizumab for PNH patients with clinically significant extravascular hemolysis. This study developed a population PK model (two-compartment with zero-order release and first-order absorption) and a PK-PD model (inhibitory E_{max}) linking danicopan exposure to alternative pathway (AP) activity. The PK model, built from 407 subjects (7491 samples), identified body weight, sex, severe renal impairment, formulation, and food status as significant covariates, though none were clinically significant. The PK-PD model estimated IC_{50} and IC_{90} for AP inhibition as 12 ng/mL and 108 ng/mL, respectively. Simulations confirmed that both 150 mg and 200 mg tid regimens achieve trough concentrations above IC_{90} and near-complete AP inhibition (< 10% residual activity) irrespective of food, supporting the approved dosing and label claims.

Scientific Context & Motivation

PNH patients on C5 inhibitors may still experience extravascular hemolysis due to C3 opsonization. Danicopan inhibits factor D, blocking the alternative pathway upstream of C5, thereby reducing EVH. Prior to this analysis, the exposure-response relationship and the impact of covariates on danicopan PK were not fully characterized. This study fills the gap by quantifying the PK-PD relationship and justifying the approved dosing regimens, demonstrating that near-

complete AP inhibition is achieved at trough with both doses, and that food has minimal impact on bioavailability.

✂ Methodological Snapshot

The population PK model was developed using NONMEM 7.5 with FOCE-I method. Data from 14 clinical studies (phase 1-3) including 407 participants (316 healthy, 91 PNH) were used. A two-compartment model with linear elimination and absorption via zero-order release followed by first-order absorption was selected. Covariates were tested via stepwise forward addition ($p < 0.01$) and backward elimination ($p < 0.001$). The PK-PD model used an inhibitory E_{max} model relating danicopan concentration to AP activity, with covariates evaluated at $p < 0.05$. Model evaluation included prediction-corrected VPCs and bootstrap. Simulations were performed to predict steady-state exposures and AP inhibition under various food conditions.

☒ Structural Model Breakdown

The PK model is a two-compartment model with linear elimination. Absorption is modeled as a zero-order release (duration D_1) followed by first-order absorption (K_a). The parameters are CL/F , V_c/F , Q/F , V_p/F , K_a , D_1 , T_{lag} , and F (bioavailability relative to reference). Covariate effects: CL/F and Q/F scaled allometrically with weight (exponent 0.645), V_c/F and V_p/F scaled with weight (exponent 0.876). Female sex reduced CL/F by 22.1%. Severe renal impairment reduced CL/F by 43.3%. Food status affected K_a , D_1 , and F . The PK-PD model is an inhibitory E_{max} model: $AP \text{ activity} = AP_0 - (AP_0 - I_{max}) \cdot C / (IC_{50} + C)$, where AP_0 is baseline AP activity, I_{max} is the minimum AP activity, IC_{50} is 12 ng/mL. Covariates on AP_0 included weight and East Asian region.

Detailed Methodological Analysis

Modeling Approach

Population PK model: two-compartment with zero-order release (D_1) followed by first-order absorption (K_a), linear elimination. PK-PD model: direct inhibitory E_{max} model linking danicopan concentration to AP activity. Software: NONMEM 7.5 (FOCE-I).

Data Sources

Data from 14 clinical studies: 11 phase 1 trials in healthy volunteers ($n = 316$), 2 phase 2 trials in PNH ($n = 22$), and 1 phase 3 ALPHA trial in PNH ($n = 69$). Total 407 participants, 7491 PK samples. AP activity data from ALPHA trial ($n = 69$, 475 time-matched samples).

Estimation Methods

First-order conditional estimation with eta-epsilon interaction (FOCE-I) in NONMEM 7.5.

Model Evaluation

Prediction-corrected visual predictive checks (pcVPC) stratified by food status; goodness-of-fit plots; bootstrap (not explicitly mentioned but typical). The pcVPC showed good prediction of peak and trough concentrations.

Covariate Analysis

Stepwise forward addition ($\alpha = 0.01$) and backward elimination ($\alpha = 0.001$) for PK model; for PD model, $\alpha = 0.05$. Covariates tested: demographics, renal/hepatic function, food status, formulation, background C5 inhibitor, etc. Significant covariates: body weight (on CL , V_c , V_p , Q), sex (on CL), severe renal impairment (on CL), food status (on K_a , D_1 , F), formulation (on F , K_a). For PD: weight and East Asian region on baseline AP activity.

Statistical Rigor Assessment

The analysis used a large dataset (407 subjects, 7491 samples) with appropriate methods (FOCE-I). Covariate selection used stringent criteria ($p < 0.01$ forward, $p < 0.001$ backward) reducing false positives. Model evaluation with pcVPC demonstrated adequate predictive performance. However, the number of patients with severe renal impairment was small ($n = 8$), limiting the precision of the covariate effect. The PK-PD model used only phase 3 data ($n = 69$) and may not capture variability in broader populations. The use of post-hoc parameters for simulations may introduce bias. Missing data handling: BLQ samples excluded (3.8%), which is acceptable.

Key Findings

Danicopan PK was described by a two-compartment model with zero-order release and first-order absorption. CL/F was 79.5 L/h, V_c/F 163 L, K_a (tablet) $1.58 h^{-1}$. Body weight, sex, severe renal impairment, formulation, and food status were significant covariates. The PK-PD model estimated $IC_{50} = 12$ ng/mL and $IC_{90} = 108$ ng/mL for AP inhibition. Simulations showed that both 150 mg and 200 mg tid achieve trough concentrations above IC_{90} and near-complete AP inhibition ($< 10\%$ residual activity) regardless of food status. The 200 mg tid regimen provided higher C_{trough} and a greater proportion of patients above IC_{90} (86.1% vs 74.5% with standard meal).

Clinical & Regulatory Implications

The results support the approved dosing of danicopan 150 mg tid, with option to escalate to 200 mg tid based on clinical response. No dose adjustment is needed for body weight, sex, mild/moderate renal impairment, or food intake. In patients with severe renal impairment, a reduced starting dose of 100 mg tid is recommended by EMA, though US and Canada labels do not require adjustment. The PK-PD relationship confirms that near-complete AP inhibition is achieved, supporting the efficacy observed in the ALPHA trial. These findings may facilitate regulatory submissions and label expansions.

Strengths & Limitations

Strengths

- Large dataset from 14 clinical studies including healthy and PNH patients.
- Rigorous covariate analysis with forward addition and backward elimination.
- Comprehensive model evaluation with prediction-corrected VPC.
- Clear linkage of PK to PD (AP activity) supporting dosing rationale.
- Inclusion of food effect assessment enabling label claim.

Limitations (Acknowledged by Authors)

- Small number of patients with severe renal impairment ($n = 8$) limits precision of covariate effect.
- PK-PD model based only on phase 3 data ($n = 69$) may not capture full variability.
- Use of post-hoc parameters for simulations may introduce bias.

Limitations (Expert Review)

- The model did not account for potential time-dependent changes in AP activity or receptor dynamics.
- The E_{max} model assumed direct inhibition without delay; no indirect response or transduction delay was considered.
- The analysis did not explore the relationship between AP inhibition and clinical endpoints like hemoglobin or transfusion avoidance.
- The covariate analysis for PD used a small sample size and may have missed subtle effects.
- The model was developed using data from clinical trials with strict inclusion criteria; generalizability to real-world PNH patients may be limited.

Generalizability

The PK model was developed using a diverse population including healthy volunteers and PNH patients across multiple regions, supporting generalizability. However, the PD model was derived from a single phase 3 trial with 69 patients, which may limit extrapolation to broader populations. The findings are most applicable to PNH patients on stable C5 inhibitor therapy with clinically significant EVH.

Key Equations

Two-compartment PK model with zero-order release and first-order absorption

$$\frac{dA_{\text{abs}}}{dt} = \begin{cases} k_0 - K_a \cdot A_{\text{abs}} & \text{if } t < D_1 \\ -K_a \cdot A_{\text{abs}} & \text{if } t \geq D_1 \end{cases} \quad \frac{dA_c}{dt} = K_a \cdot A_{\text{abs}} - \left(\frac{CL}{V_c} + \frac{Q}{V_c} \right) \cdot A_c + \frac{Q}{V_p} \cdot A_p \quad \frac{dA_p}{dt} = \frac{Q}{V_c} \cdot A_c - \frac{Q}{V_p} \cdot A_p$$

Structural PK model describing danicopan absorption (zero-order release rate k_0 for duration D_1 , then first-order absorption K_a), central compartment (A_c), and peripheral compartment (A_p). Parameters are apparent (divided by F).

Inhibitory Emax model for AP activity

$$AP = AP_0 - (AP_0 - I_{\max}) \cdot \frac{C}{IC_{50} + C}$$

PK-PD model linking danicopan concentration (C) to alternative pathway (AP) activity. AP_0 is baseline AP activity, $I_{\max}$ is the minimum AP activity at infinite concentration, IC_{50} is the concentration producing 50% of maximum inhibition.

Covariate effect on apparent clearance

$$CL/F = 79.5 \cdot \left(\frac{WT}{75} \right)^{0.645} \cdot e^{-0.221 \cdot SEX} \cdot e^{-0.433 \cdot RI} \cdot e^{\eta_{CL}}$$

Typical value of apparent clearance (L/h) with covariate effects: weight (WT in kg, centered at 75 kg), sex ($SEX = 1$ for female, 0 for male), severe renal impairment ($RI = 1$ if present, 0 otherwise). Exponent 0.645 for weight, -0.221 for female, -0.433 for severe RI .

Covariate effect on central volume of distribution

$$V_c/F = 163 \cdot \left(\frac{WT}{75} \right)^{0.876} \cdot e^{\eta_{V_c}}$$

Apparent central volume of distribution (L) with allometric scaling by weight (exponent 0.876).

Food effect on relative bioavailability

$$F = F_{\text{ref}} \cdot e^{-0.143 \cdot FASTED + 0.101 \cdot HIGHFAT}$$

Relative bioavailability (F) compared to reference (standard meal). Fasted state reduces F by 14.3% (factor $e^{-0.143}$), high-fat meal increases F by 10.1% (factor $e^{0.101}$).

Figures & Tables

- **Figure 1:** Prediction-corrected visual predictive check (pcVPC) for the final PK model stratified by food status (standard meal, fasted, high-fat meal).
 - *Significance:* Demonstrates that the model accurately predicts the central tendency and variability of danicopan concentrations across food conditions, supporting model adequacy.
- **Figure 2:** Forest plot showing the impact of covariates and dose on danicopan steady-state $C_{\max}$, C_{trough} , and AUC .
 - *Significance:* Visualizes the magnitude of covariate effects, confirming that while statistically significant, the changes are within 0.8 – 1.25 fold and not clinically meaningful.
- **Figure 3:** Simulated steady-state PK profiles and AP activity for 150 mg and 200 mg tid regimens under different food statuses.
 - *Significance:* Shows that both doses achieve trough concentrations above IC_{90} and near-complete AP inhibition regardless of food, supporting the dosing recommendations.

- **Table 1:** Baseline demographics and characteristics of the analysis populations (healthy, PNH, ALPHA study).
 - *Significance:* Provides context for the study population and highlights differences between healthy volunteers and PNH patients.
 - **Table 2:** Final population PK parameter estimates with relative standard errors and shrinkage.
 - *Significance:* Key reference for the structural model and covariate effects, showing precise estimation of CL/F , V_c/F , K_a , and covariate coefficients.
 - **Table 3:** Simulated steady-state PK exposure and AP activity for 150 mg and 200 mg tid by food status.
 - *Significance:* Quantifies the predicted C_{trough} , C_{max} , AUC , and AP inhibition, demonstrating that both doses achieve target trough concentrations above IC_{90} .
 - **Table 4:** Final PK-PD model parameter estimates (AP_0 , IC_{50} , I_{max}) with RSE and shrinkage.
 - *Significance:* Provides the estimated IC_{50} (12 ng/mL) and IC_{90} (108 ng/mL) for AP inhibition, which are critical for dose justification.
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Code & Reproducibility Assessment

No code or data availability statement provided beyond a link to Alexion’s clinical trial data sharing policy. The models were built with NONMEM 7.5, a proprietary software, and the data are not publicly available. Reproducibility is limited.

Supplementary Materials

Supplementary materials include Table S1 (study design details), Figures S1-S3 (additional diagnostic plots for the PK model), and the model code may be available upon request.

Future Directions

Future studies could explore the relationship between AP inhibition and long-term clinical outcomes, develop a disease progression model linking AP activity to hemoglobin levels and transfusion requirements, and evaluate danicopan in other complement-mediated diseases. Additionally, a more mechanistic PK-PD model incorporating factor D turnover and C3 convertase dynamics could provide deeper insights.

Expert Commentary

This work is a solid example of model-informed drug development supporting regulatory approval. The use of a zero-order release followed by first-order absorption is appropriate for danicopan’s formulation. The finding that food has minimal impact on bioavailability is clinically important and was accepted by regulators. However, the PD model’s simplicity (direct E_{max}) may not capture the full dynamics of the alternative pathway; a more mechanistic model might reveal hysteresis or tolerance. The covariate analysis is thorough, but the small number of patients with severe renal impairment warrants caution. Overall, this analysis provides a strong scientific basis for the dosing regimen and highlights the value of popPK-PD in rare disease drug development.

Bottom Line

For practicing pharmacometricians, this analysis demonstrates a robust popPK-PD approach supporting regulatory dosing. The two-compartment model with zero-order release and first-order absorption adequately describes danicopan PK, and the inhibitory E_{max} model effectively links exposure to AP inhibition. The key takeaway is that the approved 150 mg and 200 mg tid regimens achieve near-complete AP inhibition regardless of food, and that covariate effects, while statistically significant, are not clinically meaningful, except possibly in severe renal impairment where exposure increases.

☒ Figures

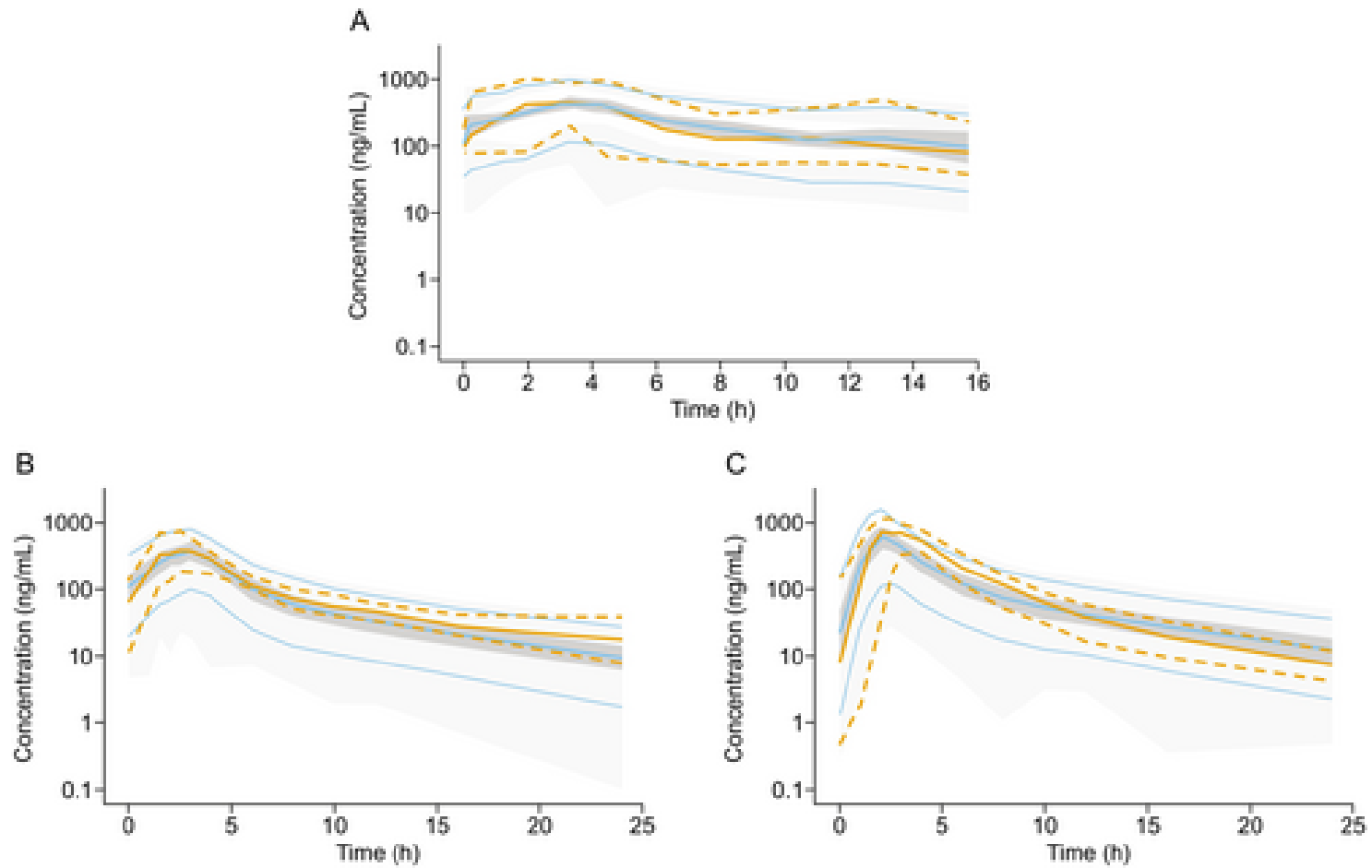


Figure 1: Final population PK model for (A) a standard meal, (B) fasted state, and (C) a high-fat meal. Orange dashed lines represent the 95th, 50th, and 5th percentiles of

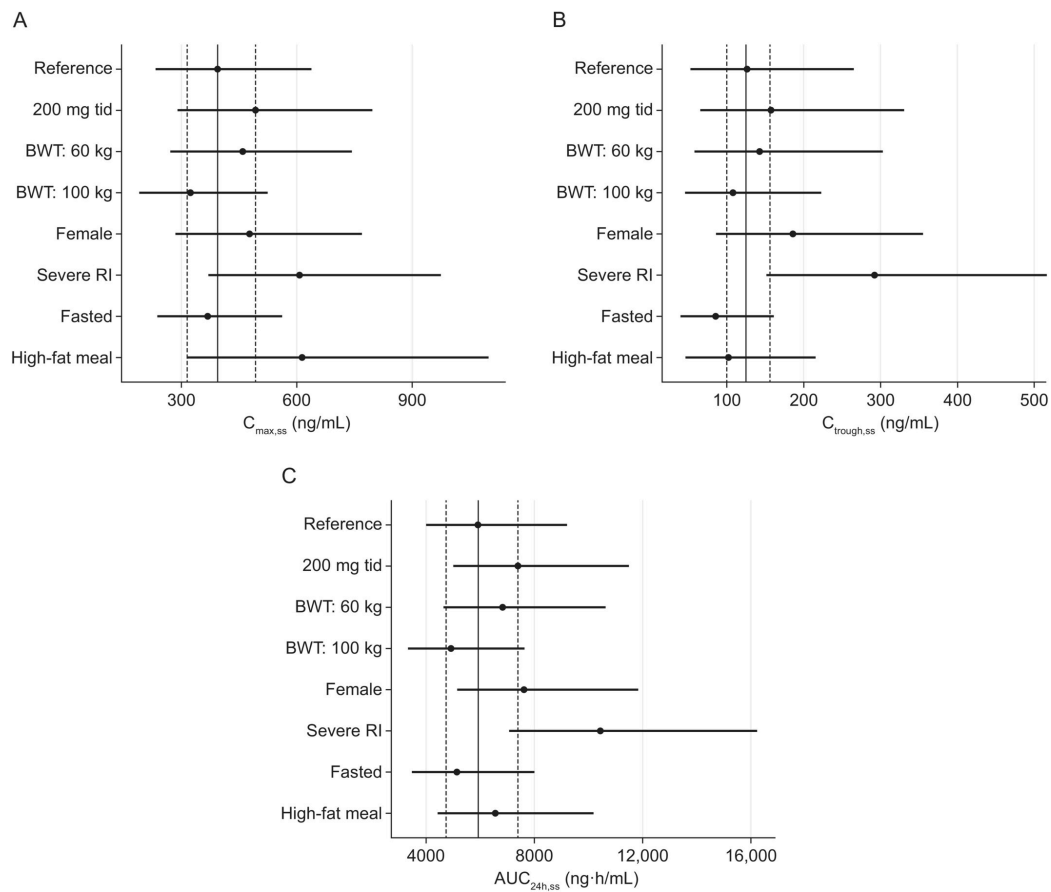


Figure 2: Impact of covariates and dose on danicopan steady state; predicted median for (A) $C_{max,ss}$, (B) $C_{trough,ss}$, and (C) $AUC_{24h,ss}$ (5-95 percentile). The solid vertical line represents the reference value.

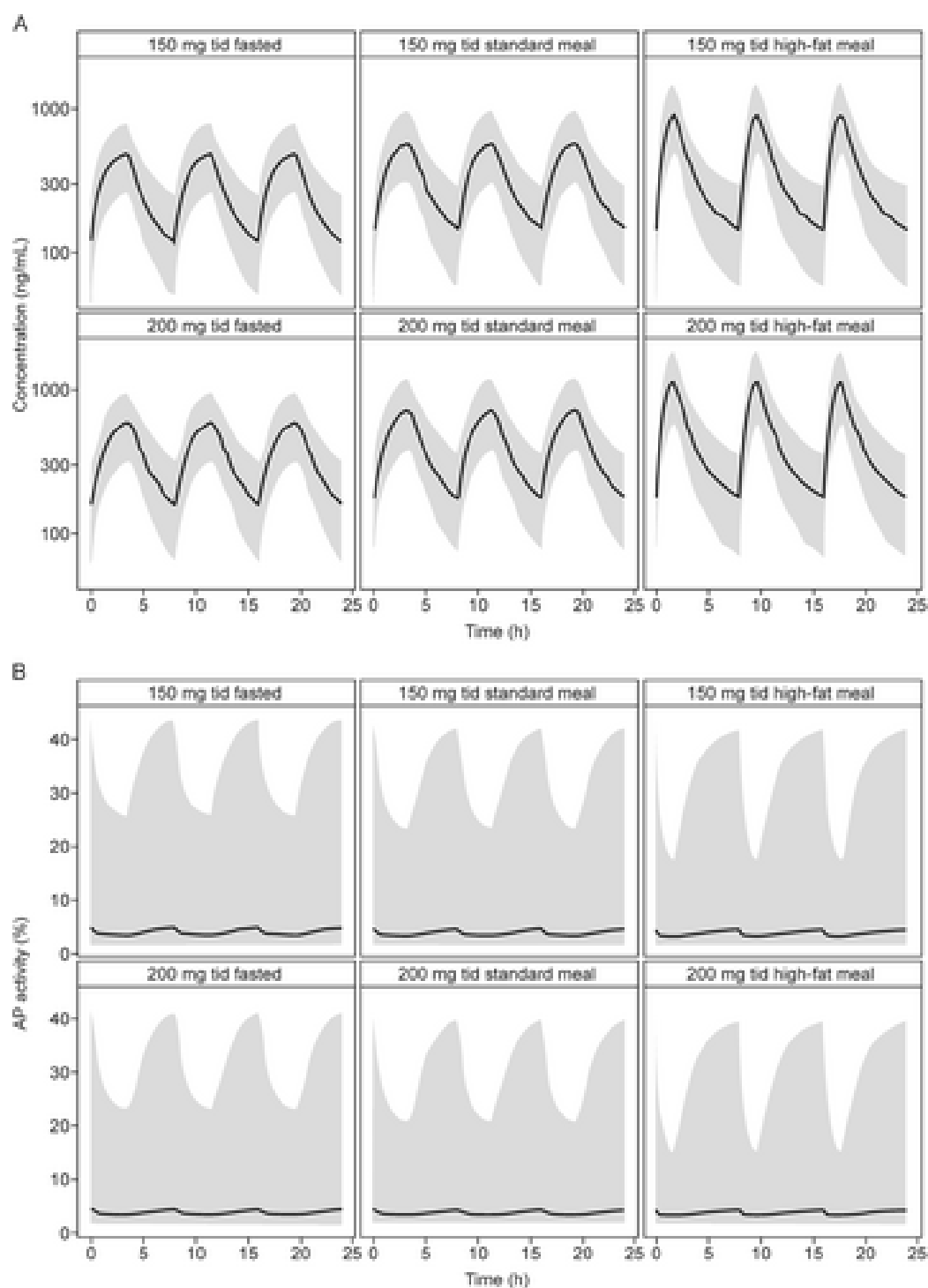


Figure 3: Simulated steady-state (A) PK and (B) AP for danicopan 150 mg and 200 mg regimens by food status. In panel A, black lines represent median values and gray region